

```
from google.colab import files
uploaded = files.upload()
```

Screenshot ... 142020.png

Screenshot 2026-07-13 142020.png(image/png) - 268920 bytes, last modified: 13/07/2026 - 100% done
Saving Screenshot 2026-07-13 142020.png to Screenshot 2026-07-13 142020.png

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

filename = list(uploaded.keys())[0]
img = cv2.imread(filename, 0)

# Laplacian mask with positive center
mask = np.array([
    [0, -1, 0],
    [-1, 5, -1],
    [0, -1, 0]
], dtype=np.float32)

sharp = cv2.filter2D(img, -1, mask)

plt.imshow(sharp, cmap='gray')
plt.title('Laplacian Sharpened Image')
plt.axis('off')
plt.show()
```

Laplacian Sharpened Image

Successfully downloaded input.jpg



Cause of Aliasing:

- Aliasing occurs when the image is sampled below the Nyquist rate.
- High-frequency details appear as jagged edges or moiré patterns.
- Quantization reduces pixel intensity precision but is not the main cause of

Corrective Approach:

- Apply an anti-aliasing (Gaussian) filter before downsampling.
- Use better interpolation methods such as `INTER_LINEAR` or `INTER_AREA`.
- Increase the sampling resolution if possible